Midterm 2 Study Sheet – Key

1. Limits.

(a) 14, by direct substitution.

(b) 1, since $\sqrt{7+2(1)}/(1+2^1) = 3/3$.

(c) 6. The initial form is $0/0$; factor $x^2-9=(x-3)(x+3)$ and cancel.

(d) $1/6$. The initial form is $0/0$; rationalizing gives $1/(\sqrt{9+t}+3)$ for $t \neq 0$.

(e) 4. The initial form is $\infty - \infty$. Rationalizing gives

$$\sqrt{x^2+8x} - x = \frac{8x}{\sqrt{x^2+8x}+x} = \frac{8}{\sqrt{1+8/x}+1} \longrightarrow 4.$$

(f) $1/4$. The initial form is $\infty - \infty$. Combine the fractions:

$$\frac{1}{x-2} - \frac{4}{x^2-4} = \frac{x-2}{(x-2)(x+2)} = \frac{1}{x+2} \longrightarrow \frac{1}{4}.$$

(g) $-\infty$, because $t-1 \to 0^-$.

(h) $5/2$, after dividing numerator and denominator by x^2.

(i) The domain excludes -1 and 2. For all other x,

$$J(x) = \frac{(x-1)(x+1)}{(x-2)(x+1)} = \frac{x-1}{x-2}.$$

The limit at -1 is $2/3$, so the graph has a hole at $(-1, 2/3)$. At 2, the left/right limits are $-\infty, +\infty$, so $x=2$ is a vertical asymptote. The limits at $\pm\infty$ equal 1, giving horizontal asymptote $y=1$. In the displayed order, the six limits are $2/3$, $-\infty$, $+\infty$, DNE, 1, 1. The two-sided limit at 2 does not exist because its one-sided limits disagree.

2. (a) Rationalizing gives $F(x) = 1/(\sqrt{x+5}+3)$ for $x \neq 4$ near 4. The limit and the required value of k are both $1/6$.

(b) The left-hand limit at 1 is 3. The right-hand limit and $B(1)$ are $2+k$, so $k=1$. Each formula is continuous elsewhere.

3. Derivatives (where the functions are differentiable).

(a) $10x^4 - \dfrac{3}{2\sqrt{x}} - \dfrac{4}{x^2}$

(b) $10(2x-1)^4$

(c) $e^{-x}(2x - x^2 - 2)$

(d) $\dfrac{2x(x+2) - (x^2+1)}{(x+2)^2}$

(e) $\dfrac{6x}{3x^2+1}$

(f) $(2x-2)e^{x^2-2x}$

(g) $\dfrac{2}{\sqrt{4x+1}}$

(h) $\dfrac{1-\ln x}{x^2}$

(i) $\dfrac{1/x + 2x}{2\sqrt{\ln x + x^2}}$

(j) $3e^{3x}(4x+5)^6 + 24e^{3x}(4x+5)^5$

(k) $28x^3 - 3$

(l) $-\dfrac{1}{x^2}$

(m) $\dfrac{1}{2\sqrt{x}}$

(n) $\dfrac{1}{x}$

(o) e^x

(p) $15(5x-2)^2$

(q) $2\ln x + \dfrac{2x+1}{x}$

(r) $\dfrac{4(1-x^2)}{(x^2+1)^2}$

(s) $\dfrac{x}{x^2+4}$

(t) $-\dfrac{e^{-2x}(2x+3)}{(x+1)^2}$

4. Tangent lines.

(a) $y' = 1/(2\sqrt{x+5})$, so the tangent line is $y - 3 = \frac{1}{6}(x-4)$.

(b) $y' = 2e^{2x}$, so the tangent line is $y = 1 + 2x$.

(c) $y' = 1/(x+1)$, so the slope at $x = e-1$ is $1/e$. The tangent line is $y - 1 = (x - e + 1)/e$, equivalently $y = (x+1)/e$.

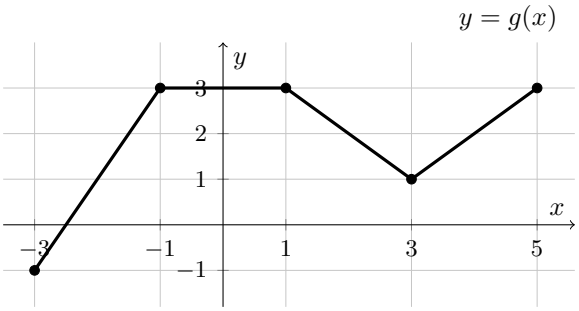
(d) $y' = [2x(x+1) - (x^2+1)]/(x+1)^2$ has value $1/2$ at 1. The line is $y - 1 = \frac{1}{2}(x-1)$.

5. (a) $F(2.9) \approx 7 + (-2)(-.1) = 7.2$.

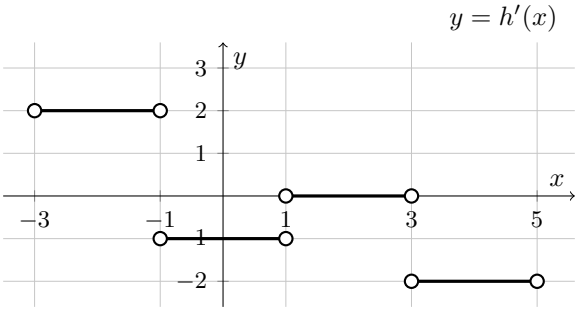
(b) $C(154) \approx 4200 + 18(4) = \$4272$. Near a production level of 150 items, total cost increases by approximately \$18 for each additional item; marginal cost is \$18/item.

(c) $R(q) = qp(q)$, so $R'(50) = p(50) + 50p'(50) = 30 - 10 = 20$ dollars/item.

6. (a) Join the points $(-3,-1), (-1,3), (1,3), (3,1), (5,3)$ with line segments.



(b) The slopes are $2, -1, 0, -2$ on $(-3,-1), (-1,1), (1,3), (3,5)$, respectively. The derivative does not exist at $-1, 1, 3$.



7. Critical numbers.

(a) $u'(x) = e^{-x}x(2-x)$, so the critical numbers are 0 and 2.

(b) The domain is $(0,\infty)$. Since $v'(x) = 1/x - 1/2 = (2-x)/(2x)$, the only critical number is 2. The derivative changes from positive to negative, giving a local maximum there. Equivalently, $v''(2) = -1/4 < 0$.

(c) $p'(x) = 3(x+3)(x-1)$ and $p''(x) = 6x+6$. The critical number -3 is a local maximum because $p''(-3) = -12$; 1 is a local minimum because $p''(1) = 12$.

(d) The critical numbers are $-2, 1, 3$. The derivative signs, in order across these numbers, are $+, -, -, +$. Thus -2 is a local maximum, 1 is a saddle point (stationary inflection), and 3 is a local minimum. For the saddle classification, $G''(x) = (x-1)(4x^2-5x-11)$ changes from positive to negative at 1. The completed sign chart is

x	$(-\infty,-2)$	-2	$(-2,1)$	1	$(1,3)$	3	$(3,\infty)$
$G'(x)$	$+$	0	$-$	0	$-$	0	$+$

8. (a) -2: local maximum; 0: local minimum (corner); 2: saddle point (stationary inflection). The signs of f' are $+, -, +, +$ on the intervals separated by these numbers.

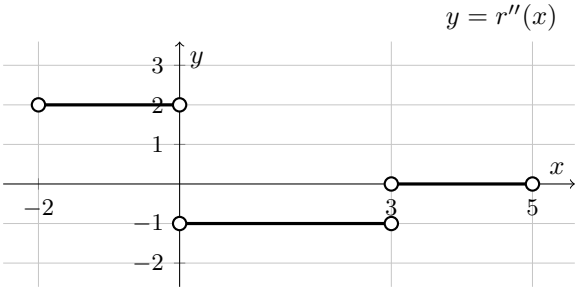
(b) The critical numbers are -1 and 2. The sign of f' is negative on $(-2,-1)$ and positive on $(-1,2)$ and $(2,4)$. Thus -1 is a local minimum; 2 is a saddle point (stationary inflection), since f' has a local minimum there and f keeps increasing.

9. Concave up on $(-4,-2)$ and $(1,3)$; concave down on $(-2,1)$ and $(3,5)$. Inflection x-coordinates: $-2, 1, 3$. These correspond to changes between increasing and decreasing behavior of p'.

10. The slopes of the three segments give

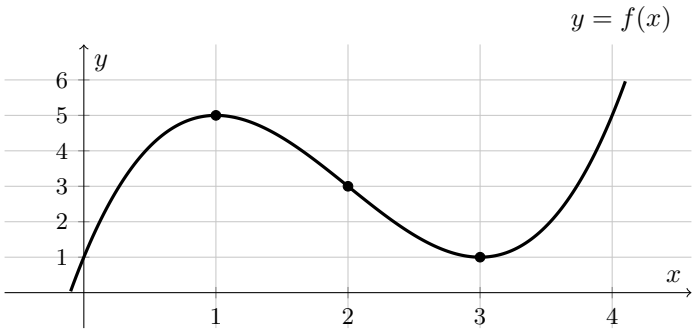
$$r''(x) = \begin{cases} 2, & -2 < x < 0, \\ -1, & 0 < x < 3, \\ 0, & 3 < x < 5. \end{cases}$$

The second derivative does not exist at 0 or 3.



11. $w''(x) = 12(x^2-1)$. Concave up on $(-\infty,-1)$ and $(1,\infty)$; concave down on $(-1,1)$. The sign changes at both boundaries, giving inflection points $(-1,-5)$ and $(1,-5)$.

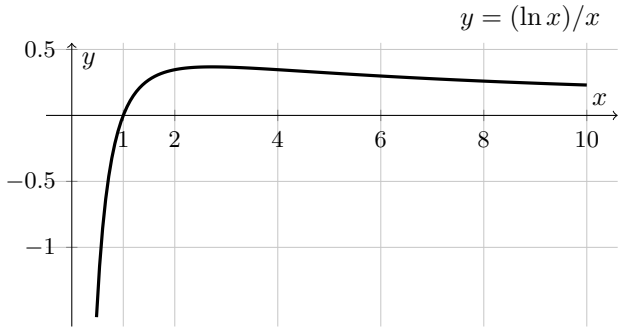
12. $f'(x) = 3(x-1)(x-3)$ and $f''(x) = 6(x-2)$. The graph increases on $(-\infty,1)$ and $(3,\infty)$, decreases on $(1,3)$, is concave down on $(-\infty,2)$ and concave up on $(2,\infty)$. Local maximum $(1,5)$; local minimum $(3,1)$; inflection point $(2,3)$.



13. The domain is $(0,\infty)$, with

$$H'(x) = \frac{1-\ln x}{x^2}, \qquad H''(x) = \frac{2\ln x - 3}{x^3}.$$

The function increases on $(0,e)$ and decreases on (e,∞), so $(e,1/e)$ is a local and absolute maximum. It is concave down on $(0,e^{3/2})$ and concave up on $(e^{3/2},\infty)$, with inflection point $(e^{3/2},3/(2e^{3/2}))$. The graph crosses the x-axis at $(1,0)$. Since $\lim_{x\to 0^+} H(x) = -\infty$ and $\lim_{x\to\infty} H(x) = 0$, its asymptotes are $x = 0$ and $y = 0$.



The limit at infinity can be justified without l'Hopital's rule: for $x \geq 1$, $\ln x = 2\ln\sqrt{x} \leq 2(\sqrt{x}-1)$, using $\ln t \leq t-1$. Therefore $0 \leq \ln x/x \leq 2/\sqrt{x} \to 0$. The inequality follows by differentiating $t - 1 - \ln t$ for $t \geq 1$.